HAOYANG WANG

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SUMMARY

M.S. student in Mechanical Engineering focusing on multimodal robotic systems, real-time perception, and intelligent control. My research background combines time-series signal processing, sensor data modeling, machine learning, and hardware-oriented system implementation. I am particularly interested in building AI-enabled robotic systems that integrate vision, audio, tactile, and physical sensor signals to improve perception, control, and interaction in real-world environments. I am currently seeking RA/PhD opportunities in multimodal robotics, embodied AI, edge intelligence, and human-robot interaction.

EDUCATION

Beijing University of Chemical Technology Major GPA: 3.54/4.0

M.S. in Mechanical Engineering (Sept.2024-Present) Ranking：2/106 Beijing, China

Related Coursework: Numerical analysis (A), Dialectics of nature (A), Engineering Testing and Signal Analysis (A-), Engineering Ethics (A-), Embedded System Virtual Instruments and Graphic Programming, Mechanical Virtual Design and Application.

B.Eng. in Process Equipment and Control Engineering (Mechanical) (Sept.2019-Jul.2023) Ranking: top 10% Beijing, China

Related Coursework: Mechanical Principles, Mechanical Design, Engineering Drawing, Chemical Principles, Process Equipment Design, C Language.

Pursued **B.A.** in Business Administration`(Sept.2023-Jul.2024) Beijing, China

Related Coursework: Production and Operations Management, Organizational Behavior, Marketing.

Honors: National Scholarship, First-Class Scholarship, Jin-Ming Encouragement Scholarship, People's Scholarship (First-Class), Merit Student.

Competitions: 1st Prize (Germination Cup, Mathematical Modeling), 2nd Prize (Internet+ Competition, 2D & 3D Engineering Graphics), Qualified of the College Student Innovation and Entrepreneurship Competition.

Leadership: Class Monitor, Debate Team Leader – School of Mechanical and Electrical Engineering, President of the Student Union of the college.

*Arts:*2nd place (Argentine tango), 5th place (Group B Modern Dance, the foxtrot) of Beijing University Sports Dance Association.

RESEARCH EXPERIENCE

Tsinghua University(Sept.2025-Present) Intern

- Developed sensor data recovery algorithms for short-term inertial sensor overload, aiming to reconstruct reliable motion signals beyond hardware dynamic range.;
- Designed and implemented control strategies for dexterous robotic hands;
- Contributed to the development and system control of intelligent desk lamps.

Multimodal Robotic Arm Research(Mar.2026-Present) Project Leader

- Built a low-cost robotic arm platform with 3D-printed structures and open-loop servo actuation;
- Currently integrating speech and vision modules for task-level interaction and grasping compensation;
- Utilizing LLMs for voice interaction and task completion, enhancing the arm's functionality with AI-driven communication.

Wind Power Generation Online Detection and Fault Diagnosis Platform(2025-Present) Researcher

- Developed a deep learning algorithm library for fault diagnosis in wind power generation;
- Built deep-learning models including CNN, CNN-LSTM, and RNN in Python to improve fault recognition accuracy and robustness.

Development of Bearing Fault Diagnosis Software(2025) Researcher

- Developed software for bearing fault detection and diagnosis, using Python's PyQt5 for UI and deep learning for fault detection;
- Designed the software to display time- and frequency-domain spectra and integrated intelligent fault detection.

Intelligent Vibration Reduction Support Design Undergraduate Design (Oct.2022-Jun.2023) Researcher

- Developed a driving module for the voice coil motor to output a certain force or displacement to counteract or suppress the vibration of the controlled object;
- Conducted Simulink and Simscape simulations and built a physical platform to test vibration reduction performance;
- Created mathematical and signal models in Matlab, and designed software using Labview to regulate system motion.

Advanced Radiation Cooling Film Technology by Electrospinning (Nov.2021-Nov.2022) Project Leader

- Investigated passive radiative cooling through electrospun thin-film fabrication and experimental thermal-performance evaluation.
- Built and tested an insulated-box setup to validate cooling effects and supported prototype exploration for automotive applications.

Dynamics Study of Liquid Droplet Impact on Thin Films(Nov.2020-Nov.2021) Project Leader

- Investigated droplet impingement on thin films, identifying critical boundary conditions and providing scientific explanations for multiphase fluid dynamics;
- Designed and built a test rig, capturing high-speed images and deriving empirical formulas for morphological changes;
- Analyzed data under varying heights and liquid thicknesses, producing results on droplet behavior and liquid crown formation.

INTERNSHIP

China Development Research Foundation(Apr.2024-Jul.2024)

- Supported research and documentation for the Ministry of Civil Affairs’ social organization evaluation by organizing institutional materials, reviewing historical records, and examining government policies and peer charitable organizations.

The supply chain startup team at Harvard Innovation Lab(Apr.2026-Present)

- Engineered a cloud-based quotation backend for a Harvard Innovation Lab startup by modularizing product size calculation and pricing rules into reusable OOP components, integrating FastAPI APIs, user authentication, file management, AWS EC2 deployment, and database-backed result storage.

ACADEMIC EXCHANGES

Academic Exchange Program, Singapore — NUS & NTU

- Engaged in faculty-led academic exchanges on technological innovation systems, AI governance, and social management. Explored how innovation policy, data governance, and institutional frameworks shape technology adoption in public and private sectors.

PUBLICATION

- Yuanhao G, Gang T* and **Haoyang W**, “ Domain Adaptation With Joint Distribution Alignment Adversarial Learning for Open-set Bearing Intelligent Fault Diagnosis ”, IEEE Sensors Journal, DOI: 10.1109/JSEN.2025.3576833.
- Guangyi Chen, **Haoyang W**, Gang T*, “TFMNet: An interpretable time-frequency mode network for mechanical equipment fault diagnosis under variable speed conditions,” Advanced Engineering Informatics, under review.
- Yuanhao G, Xi L, **Haoyang W** and Gang T*, “ Domain Adaptation With Joint Distribution Alignment Conditional Adversarial Learning for Cross-domain Intelligent Bearing Fault Diagnosis ”, revised.
- **Haoyang W**, Jianing L, Xi L, Shichao Z, Gang T*, “A Physics-Guided Multi-Domain Representation Framework for Intelligent Fault Diagnosis of Rotating Machinery”, revised.
- **Haoyang W**, Jianing L, Longjie H, Guandong F, Gang T*, “Exploring Intrinsic Fault Manifolds in Flow-Based Multimodal Representations with Temperature Information”, Working paper.
- **Haoyang W,** Jiale G* and et al.,“Inertial perception recovery beyond the hardware dynamic range under short-term overrange conditions”, Working paper.
- **Haoyang W**, Jiale G* and et al., “Audio-visual multimodal grasping compensation system for low-cost open-loop servo robotic arm”, Working paper.
- **Haoyang W**, Lidong H*, Haozhe Z, “Research on the design and intelligent control method of active-passive collaborative shock absorption system for vibration suppression”,Working paper.
- Xi L, **Haoyang W**, Gang T*, “Fault diagnosis of control moment gyroscope based on finite element simulation and cyclic generative adversarial network”, Working paper.

ADDITIONAL INFORMATION

Technical Skills: Python (PyTorch, PyQt5), MATLAB (Simulink), C, SolidWorks/Creo, CAD, Adams, LabVIEW and ANSYS.
Volunteer Work:

- Science instructor at Beijing Science Center, delivered STEM workshops for primary school children.
- Organized interactive science competitions to foster interest in mechanical engineering.

Languages: English (Proficient); Chinese (Native); Japanese (Basic).
Interests: Debate, Ballroom Dance (Award-winning), Bamboo Flute.